关于 S. Ramanujan 的一个公式

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摘要 一些有理函数的有限族具有下述性质：它们元素的幂级数展开式的系数通过一些简单的代数表达式而有关联. 我们利用 T. N. Sinha 的结果来证明一个 S. Ramanujan (拉马努金) 型的恒等式.

S. Ramanujan 的一个著名的公式说，如果

$$\frac{1+53x+9x^2}{1-82x-82x^2+x^3}=\sum_{i\geq 0}\alpha_i x^i,$$

$$\frac{2-26x-12x^2}{1-82x-82x^2+x^3}=\sum_{i\geq 0}\beta_i x^i,$$

$$\frac{2+8x-10x^2}{1-82x-82x^2+x^3}=\sum_{i\geq 0}\gamma_i x^i,$$

则

$$\alpha_i^3+\beta_i^3=\gamma_i^3+(-1)^i.$$

这个结果被 M. Hirschhorn [2] 所证明. 在 [4] 和 [3] 中给出了另外两个证明，最后一个证明是 *实例证明*, 这意味着，通过展示一个恒等式对于有限个值成立，就验证了这个恒等式 ([6], p. 9). 最近，J. McLaughlin [5] 证明了一个非常好且复杂得多的结果. [1] 中也讨论了这个结果.

在这篇短文中，我们利用 T. N. Sinha 结果的一个变形来证明下述定理.

定理 *如果*

$$\sum_{i=0}^{\infty}a_i(1)x^i=\frac{15-61x+21x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}a_i(2)x^i=\frac{13-77x-7x^2}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}a_i(3)x^i=\frac{27-135x+13x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}a_i(4)x^i=\frac{-11-49x+33x^2}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}a_i(5)x^i=\frac{17-75x-71x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}a_i(6)x^i=\frac{3-17x-91x^2}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}a_i(7)x^i=\frac{29-79x-51x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}a_i(8)x^i=\frac{-1+31x-79x^2}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}a_i(9)x^i=\frac{-13+45x-59x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}b_i(1)x^i=\frac{-1-9x+13x^2}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}b_i(2)x^i=\frac{13-67x+33x^2}{1-2x-2x^2+x^3},\qquad \sum_{i=0}^{\infty}b_i(3)x^i=\frac{-13-5x-7x^2}{1-2x-2x^2+x^3},$$

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$$\sum_{i=0}^{\infty} b_i(4)x^i = \frac{17-115x+21x^2}{1-2x-2x^2+x^3}, \qquad \sum_{i=0}^{\infty} b_i(5)x^i = \frac{29-129x+x^2}{1-2x-2x^2+x^3},$$
$$\sum_{i=0}^{\infty} b_i(6)x^i = \frac{15-81x-59x^2}{1-2x-2x^2+x^3}, \qquad \sum_{i=0}^{\infty} b_i(7)x^i = \frac{1-23x-79x^2}{1-2x-2x^2+x^3},$$
$$\sum_{i=0}^{\infty} b_i(8)x^i = \frac{3-7x-51x^2}{1-2x-2x^2+x^3}, \qquad \sum_{i=0}^{\infty} b_i(9)x^i = \frac{-11+51x-71x^2}{1-2x-2x^2+x^3},$$

以及

$$\sum_{i=0}^{\infty} b_i(10)x^i = \frac{27-35x-91x^2}{1-2x-2x^2+x^3},$$

则对于 $k=1,\ldots,8$ 和所有 $i=0,1,2,3,\ldots$ 有

$$\sum_{j=1}^{10} b_i(j)^k - \sum_{j=1}^{9} a_i(j)^k = (-1)^{ik}.$$

其证明将从两个引理而得. 在下面的引理中，我们把 $A_i(m,n)$ (和 $B_i(m,n)$) 简记为 A_i (和 B_i).

引理 1 如果

$$\begin{aligned}
&A_1 = m^2-mn-n^2, && A_2 = 15m^2-25mn-21n^2,\\
&A_3 = 13m^2-71mn+7n^2, && A_4 = 27m^2-95mn-13n^2,\\
&A_5 = -11m^2-27mn-33n^2, && A_6 = 17m^2-129mn+71n^2,\\
&A_7 = 3m^2-105mn+91n^2, && A_8 = 29m^2-101mn+51n^2,\\
&A_9 = -m^2-49mn+79n^2, && A_{10} = -13m^2-27mn+59n^2,\\
&B_1 = -m^2+3mn-13n^2, && B_2 = 13m^2-21mn-33n^2,\\
&B_3 = -13m^2-25mn+7n^2, && B_4 = 17m^2-77mn-21n^2,\\
&B_5 = 29m^2-99mn-n^2, && B_6 = 15m^2-125mn+59n^2,\\
&B_7 = m^2-101mn+79n^2, && B_8 = 3m^2-55mn+51n^2,\\
&B_9 = -11m^2-31mn+71n^2, && B_{10} = 27m^2-99mn+19n^2,
\end{aligned}$$

则对于 $k=1,\ldots,8$ 及任意的 m 和 n, 下述恒等式成立:

$$\sum_{i=1}^{10} A_i^k = \sum_{i=1}^{10} B_i^k. \tag{1}$$

其证明只是一次验算. 我们注意，这些等式是证明的核心，它们与 T. N. Sinha 的一个结果相类似 [7].

其次，我们回顾一下 Fibonacci (斐波那契) 序列，它由下面一些等式所定义:

$$F_0=0, \quad F_1=1, \quad F_{i+2}=F_{i+1}+F_i,$$

我们有公式

$$F_i = \frac{1}{\sqrt{5}}\{\alpha^i-\beta^i\},$$

其中 $\alpha=\frac{\sqrt{5}+1}{2}$, $\beta=\frac{-\sqrt{5}+1}{2}$. 提示: 注意, α,β 是 $x^2-x-1=0$ 的根. 因为 $\alpha^{i+2}-\alpha^{i+1}-\alpha^i=\alpha^i(\alpha^2-\alpha-1)=0$, 因此 α^i,β^i 都满足 Fibonacci 递推公式.

引理 2 如果 $i=0,1,2,\ldots$, 则

$$F_{i+1}^2-F_i(F_{i+1}+F_i)=F_{i+1}^2-F_iF_{i+2}=(-1)^i.$$

并且

$$\sum_{i=0}^{\infty}F_i^2x^i=\frac{x(1-x)}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}F_{i+1}^2x^i=\frac{(1-x)}{1-2x-2x^2+x^3},$$

$$\sum_{i=0}^{\infty}F_{i+1}F_ix^i=\frac{x}{1-2x-2x^2+x^3}.$$

证明 我们有

$$\begin{aligned}F_{i+1}^2-F_iF_{i+2}&=\frac{(\alpha^{i+1}-\beta^{i+1})^2}{5}-\frac{(\alpha^i-\beta^i)(\alpha^{i+2}-\beta^{i+2})}{5}\\&=\frac{-2(\alpha\beta)^{i+1}}{5}+\frac{(\alpha\beta)^i(\alpha^2+\beta^2)}{5}.\end{aligned}$$

注意到 $\alpha\beta=-1$, $\alpha^2+\beta^2=3$, 引理 2 的第 1 个恒等式从上面的等式即得.

第 1 个级数是

$$\begin{aligned}\sum_{i=0}^{\infty}F_i^2x^i=\sum_{i=0}^{\infty}\frac{(\alpha^i-\beta^i)^2}{5}x^i&=\frac{1}{5}\sum_{i=0}^{\infty}(\alpha^{2i}+\beta^{2i}-2(-1)^i)x^i\\&=\frac{1}{5}\left\{\frac{1}{1-\alpha^2x}+\frac{1}{1-\beta^2x}-\frac{2}{1+x}\right\},\end{aligned}$$

因而即得第 1 个恒等式.

用类似的方法证明其它两个级数. ■

定理的证明 在引理 1 的恒等式中，令

$$m=F_{i+1},\qquad n=F_i.$$

则

$$A_1=m^2-mn-n^2=F_{i+1}^2-F_i(F_{i+1}F_i)=F_{i+1}^2-F_iF_{i+1}=(-1)^i,$$

其中最后的等式由引理 2 得.

在 m 和 n 上面的定义下, 令 $a_i(1),\ldots,a_i(9)$ 与 $A_2,\ldots,A_{10}$ 对应, 并令 $b_i(1),\ldots,b_i(10)$ 与 $B_1,\ldots,B_{10}$ 对应，就得到定理中叙述的有理函数. 下面，我们来计算所述的第 1 个有理函数，即与 $a_i(1)$ 相应的有理函数，但是除此之外我们略去冗长的计算.

我们有

$$a_i(1)=A_2=15F_{i+1}^2-25F_{i+1}F_i-21F_i^2.$$

因而，利用引理 2 得

$$\sum_{i=0}^{\infty}a_i(1)x^i=\frac{15-61x+21x^2}{1-2x-2x^2+x^3},$$

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[25] ——, On the vanishing of and spanning sets for Poincaré series for cusp forms, Acta Math. 153 (1984), 47–116.
[26] ——, Cusp forms associated to loxodromic elements of Kleinian groups, Duke Math. J. 52 (1985), 587–625.
[27] ——, Creating an American mathematical tradition: The extended Ahlfors-Bers family, A Century of Mathematical Meetings, Amer. Math. Soc., Providence, RI, 1996, p. 265–280.
[28] R. S. Kulkarni, Some topological aspects of Kleinian groups, Amer. J. Math. 100 (1978), 897–911.
[29] A. Marden, The geometry of finitely generated Kleinian groups, Ann. of Math. 99 (1974), 383–462.
[30] C. Pommerenke, On Bloch functions, J. London Math. Soc. 2 (1970), 689–695.
[31] D. Sullivan, On the ergodic theory at infinity of an arbitrary discrete group of hyperbolic motions, Riemann Surfaces and Related Topics, Ann. of Math. Stud., vol 97, Princeton Univ. Press, Princeton, NJ, 1980, p. 465–496.
[32] ——, A finiteness theorem for cusps, Acta Math. 147 (1981), 289–299.
[33] ——, Quasiconformal homeomorphisms and dynamics I, Solution of the Fatou-Julia problem on wandering domains, Ann. of Math. 122 (1985), 401–418.
[34] W. P. Thurston, Three dimensional manifolds, Kleinian groups and hyperbolic geometry, Bull. Amer. Math. Soc. (N.S.) 6 (1982), 357–381.
[35] H. Yanagihara, On the locally univalent Bolch constant, J. Analyse Math. 65 (1995), 1–17.
[36] Commentationes in honorem Lars V. Ahlfors, LXXX Annos NATO, Ann. Acad. Sci. Fenn. Ser. A I Math. Dissertationes 13 (1988).

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正如所述.

现在，定理从恒等式 (1) 即得. ■

致谢 (略)

参考文献

[1] G. Andrews, B. Berndt, Ramanujan's Lost Notebook: Part IV. Springer-Verlag, New York, 2013.
[2] M. D. Hirschhorn, An amazing identity of Ramanujan, Math. Mag. 68 no. 3 (1995) 199–201, `http://dx.doi.org/10.2307/2691416`.
[3] M. D. Hirschhorn, A proof in the spirit of Zeilberger of an amazing identity of Ramanujan, Math. Mag. 69 (1996) 267–269, `http://dx.doi.org/10.2307/2690530`.
[4] M. D. Hirschhorn, J. H. Han, Another look at an amazing identity of Ramanujan, Math. Mag. 79 no. 4 (2006) 302–304, `http://dx.doi.org/10.2307/27642956`.
[5] J. McLaughlin, An identity motivated by an amazing identity of Ramanujan, Fibonacci Quart. 48 (2010) 34–38.
[6] M. Petkovšek, H. S. Wilf, D. Zeilberger, $A = B$. A K Peters/CRC Press, Wellesley, MA, 1996.
[7] T. N. Sinha, On the Tarry-Escott problem, Amer. Math. Monthly 73 no. 3 (1966) 280–285, `http://dx.doi.org/10.2307/2315345`.

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